VibeScan: Multi-Dimensional Benchmark Analysis for JavaScript Security Scanning

A reproducible multi-dimensional study on adjudicated DVNA with aligned peer tool runs: recall, cases-per-alert precision proxy, per-family coverage, and wall-clock scan time. CCSC 2026.

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Research Questions

CROSS-TOOL DVNA RESULTS

Tools: 1 VibeScan 2 Bearer 3 Snyk Code 4 Semgrep 5 CodeQL 6 ESLint

FIGURE 3 — COVERAGE BY VULNERABILITY CATEGORY (DVNA CORE)

VibeScan is the only tool with full coverage across all vulnerability categories, especially outperforming in complex injection and client-side cases.

Category · Scenarios covered		1 VibeScan	2 Bearer	3 Snyk	4 Semgrep	5 CodeQL	6 ESLint
Injection	SQL, Command, NoSQL (3)	100%	100%	66%	33%	66%	33%
Redirect & Nav	Open Redirect (1)	100%	100%	100%	100%	100%	0%
Integrity	Unsafe Deserialization (1)	100%	0%	100%	100%	100%	0%
Secrets & Crypto	Secrets + Weak Hash (2)	100%	100%	100%	50%	50%	0%
Auth & Client	Env + XSS (2)	100%	0%	50%	0%	50%	0%
Overall recall (11 cases)		100%	72.7%	63.6%	36.4%	54.5%	9.1%

The matrix groups DVNA's 11 benchmark cases into five vulnerability categories. VibeScan is the only scanner to achieve 100% coverage across all five. Peer tools perform well on Injection and Redirect but drop sharply in Auth & Client-side, where only VibeScan detects both cases — highlighting the coverage gap that raw alert counts obscure.

eslint-plugin-security-shape.bone : 1/115 recall : 493 alerts

2. Taint / Data Flow
Source-to-sink tracking within files.

FIGURE 2 — RECALL VS. RAW ALERT VOLUME

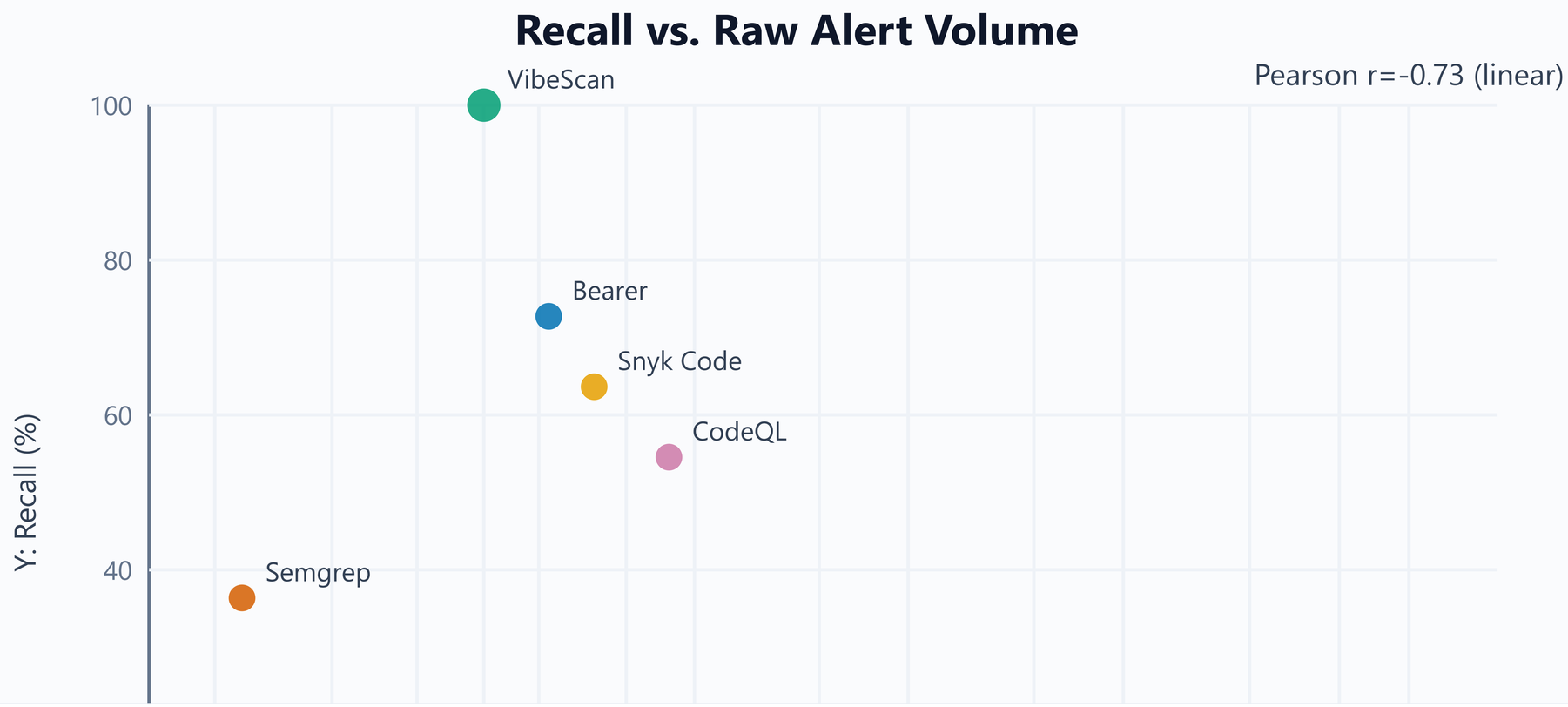
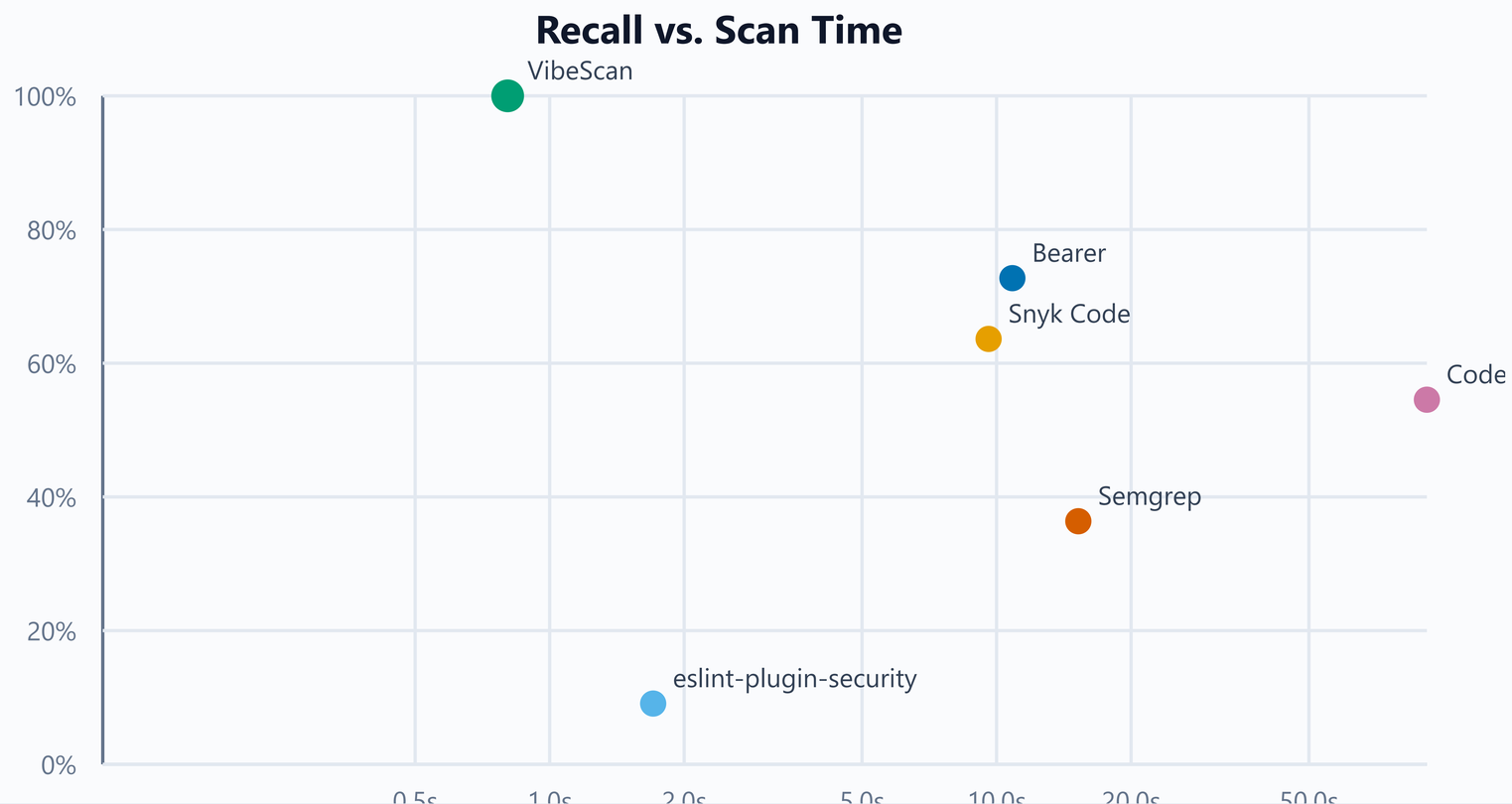
FIGURE 5 — RECALL VS. SCAN TIME